Topic: Step functions

Question: Represent $f(t)$ in terms of unit step functions.

$$f(t) = \begin{cases} 5 & 0 \leq t < 2 \\ 2 & 2 \leq t < 3 \\ 7 & 3 \leq t < 5 \\ 0 & t \geq 5 \end{cases}$$

Answer choices:

A $f(t) = 5 - 3u_2(t) + 5u_3(t) - 7u_5(t)$

B $f(t) = 5 + 3u_2(t) - 5u_3(t) + 7u_5(t)$

C $f(t) = 5 - 3u_2(t) + 5u_3(t)$

D $f(t) = 5u_2(t) + 2u_3(t) + 7u_5(t)$

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Solution: A

Remember that a Heaviside function represents a switch, so we need one Heaviside function every time we make a switch in the value of $f(t)$. There are three switches, one from 5 to 2, one from 2 to 7, and one from 7 to 0, so we'll need three Heaviside functions in order to express $f(t)$.

We write a Heaviside function as $u_c(t)$, where the switch occurs at $t = c$. The switches in this function $f(t)$ are at $t = 2$, $t = 3$, and $t = 5$, which means the three Heaviside functions we'll need are $u_2(t)$, $u_3(t)$, and $u_5(t)$.

To express $f(t)$, we start with the idea that $f(t)$ has a value of 5 when it's "off." When the first switch at $t = 2$ turns on, the value of the function immediately shifts from 5 to 2 (down 3 units), so we can write

$$f(t) \approx 5 - 3u_2(t)$$

We use -3 as a coefficient on $u_2(t)$, since $5 - 2 = 3$, and 2 is the value the function takes on after the first switch. Then moving from 2 to 7 is an increase of 5, which means we need to multiply $u_3(t)$ by 5,

$$f(t) \approx 5 - 3u_2(t) + 5u_3(t)$$

and moving from 7 to 0 is a decrease of 7, which means we need to multiply $u_5(t)$ by -7, and this will give us the full expression of $f(t)$ in terms of step functions.

$$f(t) \approx 5 - 3u_2(t) + 5u_3(t) - 7u_5(t)$$

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Topic: Step functions

Question: Which function has an (on at 7)-to-off switch at $t = 10$?

Answer choices:

A $f(t) = 10 - 10u_7(t)$

B $f(t) = 10u_7(t)$

C $f(t) = 7u_{10}(t)$

D $f(t) = 7 - 7u_{10}(t)$

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Solution: D

An (on at n)-to-off switch can be modeled by

$$f(t) = n(1 - u_c(t)) = \begin{cases} n & 0 \leq t < c \\ 0 & t \geq c \end{cases}$$

If we want the switch to be "on" at a value of 7, and then turn "off" to a value of 0 when we arrive at the switching point $t = 10$, then we'd write the function as

$$f(t) = 7(1 - u_{10}(t)) = \begin{cases} 7 & 0 \leq t < 10 \\ 0 & t \geq 10 \end{cases}$$

$$f(t) = 7 - 7u_{10}(t) = \begin{cases} 7 & 0 \leq t < 10 \\ 0 & t \geq 10 \end{cases}$$

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Topic: Step functions

Question: How many switches does the function have?

$$f(t) = 9 + 2u_2(t) - 3u_5(t)$$

Answer choices:

A 1

B 2

C 3

D 5

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Solution: B

The unit step functions show us that we have switches at $t = 2$ and $t = 5$.

The first term, 9, just gives us the "starting point" of the function.

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